Journal of Experimental Criminology (2025) 21:915–938
https://doi.org/10.1007/s11292-024-09613-0

Crime control or just theater? An experimental test of the effects of a mobile safety app on crime prevention intentions and behaviors

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Accepted: 22 January 2024 / Published online: 3 February 2024

Abstract

Objectives We examined whether mobile safety apps constitute a form of crime control theater that thwarts actions that have the potential to prevent crime.

Methods Using an experimental research design with a sample of college students ($N = 210$), we measured the effects of a safety app (i.e., Rave Guardian) on trust in mobile technology, fear of crime victimization, self-protective behavior, and bystander intentions to help someone at risk of victimization.

Results Findings indicate that safety app access is associated with increased self-protective behavior and bystander intentions. However, participants who actually used the app exhibited increased trust in mobile technology that slightly inhibited bystander intentions. We found no effect of app access or usage on fear of victimization.

Conclusions Safety app usage, but not access, may function as a form of crime control theater. Until future research further explores differences between app access and usage, widespread uptake seems premature.

Keywords Bystander · Crime control theater · Fear · Mobile safety app · Prevention · Trust · Campus victimization

Young Americans place a great deal of trust in mobile devices, and they tend to believe that these devices can keep them safe from crime (Cumiskey and Brewster

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2012; Nasar et al., 2007). Building on this trust, mobile application developers have released hundreds of safety apps on the market with the purported aim of keeping users safe from victimization (Bivens and Hasinoff, 2018; Ford et al., 2022; Wood et al., 2022). Yet, there is a dearth of research exploring the effectiveness of safety apps that are designed for primary crime prevention, which aims to "modify or abate the social or environmental conditions that give rise to crime or allow potential victims to manage their exposure to risky environments or circumstances" (Wood et al., 2022, p. 1096). That is, there is no empirical evidence that mobile safety apps are effective in preventing crime victimization among general populations of users (Cardoso et al., 2019; Maxwell et al., 2020; Wood et al., 2022).

This lack of evidence has led some scholars to claim that mobile safety apps constitute a form of what Griffin and Miller (2008) call "crime control theater," in that they create the mere illusion of crime control without actually preventing crime (McMillan and White 2021). One key characteristic of crime control theater interventions is that they often have unintended consequences that may be counterproductive to their overall goal of crime prevention (DeVault et al., 2016). As such, scholars have proposed that mobile safety apps may carry the unintended consequence of reducing fear of crime, which may then lead users to engage in fewer safety behaviors that may protect them from victimization (Bivens and Hasinoff 2018; Hasinoff 2017; Maxwell et al., 2020). Additionally, trust in mobile safety apps may amplify what Darley and Latane (1968) have called the "bystander effect," which entails the diffusion of responsibility for helping in group settings. Specifically, users who place trust in mobile technology may be less inclined to offer help to someone who is at risk of crime victimization (Kettrey and Thompson 2022).

In this study, we tested the general proposition that mobile safety apps constitute a form of crime control theater which can thwart actions that have the potential to prevent crime. Using an experimental research design and path analysis modeling, we examined the effects of one mobile safety app (i.e., Rave Guardian) on trust in mobile technology, fear of crime victimization, self-protective behavior, and bystander intentions to help someone who is at risk of crime victimization.

Mobile safety apps as crime control theater

Crime control theater is "a public response or set of responses to crime which generate the appearance, but not the fact, of crime control" (Griffin and Miller 2008, p. 160). Griffin and Miller argue that crime control theater is ineffective at preventing crime and is often tied to moral panics, such that it frequently represents an "effort to socially construct a solution to a socially constructed problem" (p. 160). Examples of crime control theater include AMBER alerts and sex offender registration laws, each of which has been empirically demonstrated to be ineffective at preventing an infrequent problem. For example, AMBER alerts were designed as lifesaving efforts to recover abducted children, yet analyses of AMBER alert cases indicate that most abductions are not life-threatening and child recovery is predicted by the child's relationship to the abductor rather than by the issuance of an alert (Griffin et al., 2022; Griffin et al., 2016). Similarly, sex offender registries were designed to prevent sexual recidivism, but meta-analyses have indicated that recidivism is extremely rare among sex offenders and is not impacted by registration (Kim et al., 2016; Zgoba and Mitchell 2023). Yet, despite the ineffectiveness of sex offender registries, the general public tends to believe that they are useful and that they keep communities safe (Anderson and Sample 2008; Harris and Cudmore 2018; Levenson et al., 2007).

Importantly, DeVault et al., (2016) note that crime control theater is not only ineffective but may also be associated with unintended consequences. For example, research indicates that property values fall in neighborhoods where registered sex offenders reside, as homebuyers are presumably less inclined to purchase homes in these areas (Caudill et al., 2015; Linden and Rockoff 2008; Pope 2008; Yeh 2015).

Mobile safety apps arguably constitute crime control theater for two reasons. First, they feed into a panic about stranger attacks in public spaces, which are much rarer than crimes committed by acquaintances (Cardoso et al. 2019; Maxwell et al., 2020). Second, although safety apps have flooded the market in recent years, there is no evidence that mobile safety apps are effective at preventing crime. Instead, they may "offer the *illusion* of safety rather than *actual* safety" (McMillan and White 2021, p. 474 [emphasis in original]).

Recent reviews of mobile safety apps note that hundreds are currently available, with a wide range of features. These include crime mapping/reporting that allows users to receive/contribute notifications of crime in their area, alerting personal contacts or authorities when a user is in danger, monitoring the user's environment for deviations from a designated path home or for increases in ambient noise, recording photo/audio/video evidence, allowing the user to take evasive action (e.g., sounding an alarm), or providing informational resources such as safety tips and referrals to services (Bivens and Hasinoff 2018; Ford et al. 2022; Wood et al. 2022). Most of these apps are marketed to women as a means of reducing their personal risk of sexual and physical assault, with little evidence of their effectiveness in preventing such crime (Cardoso et al. 2019; Maxwell et al., 2020). In fact, a scoping review of the research on primary crime prevention apps concluded that most existing studies analyze the design innovations of these apps without systematically evaluating their effectiveness of preventing crime (Wood et al. 2022). Furthermore, findings from a quasi-experimental study that is purportedly the first to examine effects of a primary crime prevention app found no effects on users' victimization (Capellan et al. 2022).

While there is a dearth of evidence demonstrating effects of mobile safety apps on their intended outcomes, some scholars have noted the possibility of one important unintended consequence of these apps: reduced fear of crime (Bivens and Hasinoff 2018; Hasinoff 2017; Maxwell et al., 2020). Reduced fear of crime could be counterproductive to crime prevention. This is because fear of crime is associated with higher levels of self-protective behavior (Liska et al. 1988; Rader et al. 2009). Thus, reductions in fear of crime may be associated with reductions in self-protective behavior. Applied specifically to mobile safety apps, Maxwell et al. explain:

Although personal safety apps may contribute to a reduction in FOC [fear of crime], the fact that users may have a reduced FOC without any reduction in their actual vulnerability to victimization may be problematic. . . A false sense of security engendered by use of a personal safety app may result in greater risk-taking behavior (p. 243).

Despite the lack of evidence for crime prevention, and the potential for unintended consequences, support for personal safety apps is high. Users tend to rate these safety apps positively and view them as useful (Ford et al. 2022). This is not necessarily surprising considering the ubiquity and high levels of trust placed in mobile technology.

Trust in mobile safety apps and a technology-amplified bystander effect

Mobile phones are ubiquitous, and young adults place a great deal of trust in mobile technology. Data from the Pew Research Center indicate that 96% of young adults aged 18 to 29 own a smart phone (Pew Research Center 2019). Additionally, young people report that having a mobile phone makes them feel safer at night and, with a mobile phone, they walk in places where they normally would not (Nasar et al. 2007). In fact, the authors of one randomized study asked female college students how safe they would feel in a public space with either pepper spray or a mobile phone (Cumiskey and Brewster 2012). Participants in the mobile phone condition reported significantly higher ratings of perceived safety than those reported by participants in the pepper spray condition. This trust appears to extend to mobile safety apps, as one study found that higher education faculty, staff, and students believed the value of these apps to be three times higher than the actual cost for universities to adopt them (Yabe 2017).

The widespread use of, and trust in, mobile technology has led scholars to hypothesize that mobile safety apps could amplify a well-documented bystander effect (Kettrey and Thompson 2022). This hypothesis is grounded in two assumptions. First, according to Latane and Darley's (1969) foundational work on the bystander effect, bystanders must meet the following criteria in order to intervene in an emergency situation: notice the event, define the event as warranting action/intervention, take responsibility for acting (i.e., feel a sense of personal duty), and demonstrate a sufficient level of self-efficacy (i.e., perceived competence to successfully intervene). Users who place a high level of trust in safety apps may not meet all these criteria. Specifically, their trust in the presumed utility of mobile apps may inhibit them from feeling a personal responsibility or duty to intervene.

Second, the likelihood of bystander intervention in emergency situations decreases as the number of witnesses increases (Darley and Latane 1968). Thus, individuals who witness warning signs of a crime may be less inclined to take preventative action if they have come to trust mobile technology, and they believe that potential victims have the ability to use mobile apps to request help. In other words, uptake of mobile safety apps may generalize the perceived number of witnesses to include those who are physically present as well as those who may be contacted via mobile technology. This could, in turn, amplify the diffusion of responsibility associated with the bystander effect and result in less frequent intervention behavior.

The present study

In the present study, we used an experimental research design to examine potential unintended effects of primary crime prevention apps. We specifically tested the following hypotheses:

H1:Participants randomized to a mobile app condition will exhibit lower levels of self-protective behavior than participants randomized to a control condition.

H2: Participants randomized to a mobile app condition will exhibit lower fear of crime than participants randomized to a control condition. Fear of crime will, in turn, be associated with higher levels of self-protective behavior. This will result in a negative, indirect effect of mobile safety apps on self-protective behavior.

H3: Participants randomized to a mobile app condition will exhibit lower bystander intervention intentions than participants randomized to a control condition.

H4: Participants randomized to a mobile app condition will exhibit greater trust in mobile technology than participants randomized to a control condition. Trust in mobile technology will, in turn, be associated with lower bystander intervention intentions and lower levels of self-protective behavior. This will result in a negative, indirect effect of mobile safety apps on both bystander intentions and self-protective behavior.

Materials and method

The purpose of this study was to experimentally test the aforementioned hypotheses regarding relationships between mobile safety apps, fear of crime victimization, trust in mobile technology, bystander intentions, and self-protective behaviors. The sample for the study consisted of students enrolled at a large public land-grant university in the Southeastern United States who were recruited through introductory sociology courses during the Spring 2023 semester.

Mobile safety app

In this study, we used Rave Guardian to test our hypotheses about mobile safety apps. Rave Guardian is a "custom-branded personal safety app that helps higher education institutions, businesses, and healthcare organizations connect and engage with their communities wherever they are" (Rave Mobile Safety 2023). Specific app features include a "virtual escort" that allows users to notify pre-specified contacts when they reach a destination, a confidential text feature for reporting suspicious activity to local security, geo-targeted notifications that alert users to dangerous activities in their area, and confidential texting that permits users to report suspicious activity. Organizations, such as colleges and universities, may purchase subscriptions to the app and customize it to meet their institutional needs.

The university where we recruited study participants has a subscription to Rave Guardian, which students may download and use for free. The university's customizations include alerting users to crime/danger on campus and providing quick links to local resources such as campus police, parking services, bus transit, and student counseling. Results from a campus climate survey administered at our recruitment institution during the same year we conducted our study indicated that less than 10% of students had downloaded the Rave Guardian app. Thus, we believe that this specific mobile safety app was ideal for our study because it had been customized to meet the needs of our study population, and since few students had accessed the app, the threat of study contamination was relatively low.

Recruitment

To recruit participants for our study, the lead study author approached instructors of four introductory sociology courses, requesting access to their classes to recruit research participants. The research team used cluster randomization to assign participants to treatment and control groups by class. We matched the four classes by size (i.e., two classes of approximately 50 students and two of approximately 150 students) and randomized one class from each matched set to the treatment group via computer algorithm. This resulted in one of the smaller classes and one of the larger classes being randomized to the treatment group, and the remaining two classes being randomized to the control group.

Groups of three to four graduate student team members visited the four recruitment classes and read a script inviting those students in attendance to participate in a study about "mobile apps and campus safety." All classes were told that participation was optional and that they would be asked to complete a pre-survey on the day of recruitment and a post-survey about a month later. The two treatment group classes were told that they must have a smart phone with the capability of downloading mobile apps to participate. This inclusion criterion did not apply to control group classes.

The graduate student team members provided both a hyperlink and QR code where students could access the pre-survey, which was administered via Qualtrics. By clicking on the link or scanning the QR code, students were led to a study consent form and then to the survey items. For the treatment group classes, graduate student team members waited until students indicated that they were finished with the survey and then provided a hyperlink and QR code that participants used to download the mobile safety app, asking them to download and use the app for a month. Team members walked around the room, offering assistance with downloading the app, and left when students appeared to no longer have questions.

No additional instructions were provided to the control group classes after completing the pre-survey. In order to accurately track participants through the study and classify them in the appropriate class cluster, each class had its own unique link to the study survey. Once team members left the classrooms, they closed the link to the appropriate survey to prevent anyone who did not attend the recruitment session from accessing the survey, as well as to prohibit participants from taking the pre-survey after they downloaded and used the app. The recruitment process took about 10–15 min for control group classes and 15–20 min for treatment group classes.

Approximately 2 weeks after recruitment, we sent an email to all participants in the treatment group, reminding them to download the mobile safety app (providing the app link again) and reiterating that the research team would return to their class in a couple of weeks to administer a survey asking them about their experiences with the app. The control group received no correspondences from the research team between pre- and post-survey administration.

Approximately 4 weeks after recruitment, graduate student team members returned to all four recruitment classes asking students to complete a post-survey. Team members read from a pre-approved script that reminded participants of the purpose of the study and informed all students in attendance that they must have completed the pre-survey to be eligible to complete the post-survey (students who completed a post-survey without completing a pre-survey were excluded from the study). Class attendance was noticeably reduced when administering the post-survey, which was during the final weeks of the semester. Thus, we emailed all students who had participated in the pre-survey but did not complete a post-survey, providing them with a link to the appropriate survey for their class. We sent these emails immediately after the post-survey administration and followed up with reminders for up to 1 week.

Participants received a $15 Amazon gift card for submitting each survey, whether they completed all items or not. Those who only submitted a pre-survey received a single gift card, and those who submitted both pre- and post-surveys received two gift cards. We delivered electronic gift cards to participants' university email accounts, and we used university email addresses to link individual participants' pre- and post-survey responses. Since we used email addresses to link responses, our recruitment script and consent form both stated that participants must provide their university email address in order to receive a gift card for their pre- and post-survey, respectively. Procedures were reviewed and approved by [Clemson] University's Institutional Review Board.

Participants

The flow of participants through the study is summarized in the CONSORT diagram in Fig. 1. On the date when graduate student team members visited recruitment classes, there were 291 students in attendance. Of these 291 students, 287 submitted a pre-survey (122 control condition; 165 treatment condition), yielding a 98.63% response rate. Of these 291 participants, 216 completed the post-test, with 193 participants (n = 110 treatment; n = 83 control) completing the survey during in-class data collection and 23 (n = 12 treatment; n = 11 control) completing it after receiving a follow-up email. We deemed six post-surveys ineligible because their pre-survey indicated that they had downloaded Rave Guardian prior to the study (see description of measures below). Thus, we calculated the final retention rate by using 210 as the final sample size and subtracting six from the denominator (210/281), indicating that we retained 74.73% of participants between the pre-survey and post-survey.

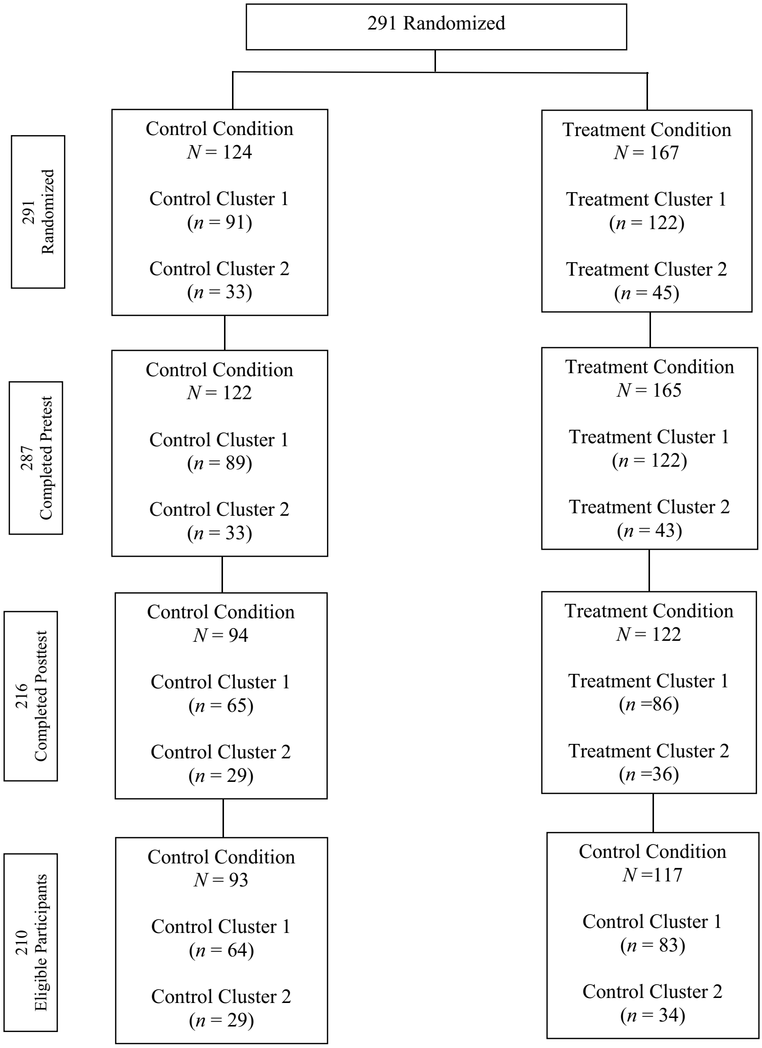

Fig. 1 CONSORT diagram depicting flow of participants through study

Table 1 provides demographic information for the final sample of participants (N = 210) for both the aggregate sample, as well as broken down by treatment condition. The aggregate sample consisted mostly of cisgender women (63.25%). The majority of participants were straight (90.60%), White (82.91%), non-Latinx (91.45%), freshmen (62.39%), and lived on campus (65.81%). Chi-squared tests of independence indicated that participants who completed the post-survey did not significantly differ ($p > .05$) from those who did not complete the post-survey (drop-outs) on any of the aforementioned variables. Additionally, the attrition rate was not significantly different between participants assigned to the treatment group and those assigned to the control group.

Table 1 Sample demographics and outcomes for eligible participants ($N = 210$)

	Safety app group			Control group	Total
	TOT ($n = 117$)	Downloaded app ITT ($n = 106$)	Used app ITT ($n = 67$)	($n = 93$)	($N = 210$)
	M (SD) or %	M (SD) or %	M (SD) or %	M (SD) or %	M (SD) or %
Demographics					
Gender					
Cisgender woman	63.25	63.21	61.19	53.76	59.05
Cisgender man	35.90	35.85	37.31	44.09	39.52
Transgender/gender non-binary	0.85	0.94	1.50	2.15	1.43
Sexual identity					
Straight	90.60	91.51	89.56	92.47	91.43
Sexual minority	9.40	8.49	10.44	7.53	8.57
Race					
White	82.91	81.13	79.10	80.65	81.90
Black	7.69	8.49	8.96	10.75	9.05
Asian	5.13	5.66	4.48	3.23	4.29
Multiracial	4.27	4.72	7.46	5.37	4.76
Latinx ethnicity	8.55	6.60	5.97	8.60	8.57
Year in school					
Freshman	62.39	61.32	65.67	68.82	65.24
Sophomore	24.79	25.47	23.88	20.43	22.86
Junior	7.69	7.55	4.48	7.53	7.62
Senior	5.13	5.66	5.97	3.22	4.28
Live on campus	65.81	66.04	67.16	72.04	68.57

Table 1 (continued)

	Safety app group			Control group	Total
	TOT ($n = 117$)	Downloaded app ITT ($n = 106$)	Used app ITT ($n = 67$)	($n = 93$)	($N = 210$)
Endogenous variable pre-survey scores					
Trust in mobile technology score	3.01 (0.81)	3.04 (0.83)	3.00 (0.81)	3.12 (0.75)	3.06 (0.78)
Negative consequences of apps	3.04 (1.10)	3.07 (1.13)	3.07 (1.11)	3.02 (1.00)	3.03 (1.06)
Cautious when using safety apps	2.64 (1.14)	2.66 (1.16)	2.55 (1.15)	2.91 (1.08)	2.76 (1.12)
Safety apps are risky	3.36 (0.87)	3.40 (0.88)	3.37 (0.90)	3.43 (0.88)	3.39 (0.87)
Fear of crime victimization score	1.63 (0.75)	1.64 (0.76)	1.71 (0.84)	1.49 (0.56)	1.57 (0.68)
Theft—day	1.47 (0.92)	1.46 (0.90)	1.51 (0.98)	1.34 (0.76)	1.41 (0.86)
Non-sexual assault—day	1.32 (0.77)	1.35 (0.81)	1.46 (0.93)	1.27 (0.65)	1.30 (0.72)
Sexual assault—day	1.30 (0.73)	1.32 (0.76)	1.45 (0.89)	1.29 (0.65)	1.30 (0.70)
Theft—night	1.90 (1.17)	1.90 (1.15)	1.96 (1.22)	1.73 (0.92)	1.82 (1.07)
Non-sexual assault—night	1.81 (1.11)	1.83 (1.13)	1.88 (1.15)	1.60 (0.84)	1.72 (1.00)
Sexual assault—night	1.97 (1.24)	1.97 (1.24)	2.03 (1.29)	1.69 (0.99)	1.85 (1.14)
Bystander intentions score	4.02 (0.64)	4.00 (0.65)	4.13 (0.58)	4.04 (0.61)	4.03 (0.63)
Watching a friend's belongings	4.53 (0.75)	4.52 (0.77)	4.66 (0.59)	4.50 (0.63)	4.52 (0.70)
Ensuring friend's safety after drinking	4.82 (0.52)	4.81 (0.53)	4.81 (0.58)	4.88 (0.36)	4.85 (0.45)
Walk with friend	4.13 (1.07)	4.09 (1.10)	4.19 (1.06)	4.07 (1.09)	4.11 (1.08)
Text/message friend	4.28 (1.03)	4.24 (1.05)	4.29 (0.99)	4.25 (1.05)	4.27 (1.03)
Watching a student's belongings	3.80 (1.13)	3.76 (1.13)	4.12 (0.96)	3.81 (1.07)	3.80 (1.10)
Ensuring student's safety after drinking	3.90 (1.04)	3.89 (1.06)	4.01 (0.93)	3.87 (0.87)	3.89 (0.97)
Walk with student	2.70 (1.15)	2.68 (1.15)	2.85 (1.14)	2.86 (1.13)	2.77 (1.14)
Self-protective behavior score	1.98 (0.92)	1.96 (0.91)	2.06 (0.95)	1.92 (0.85)	1.95 (0.88)
Avoid going outside alone at night	2.36 (1.45)	2.36 (1.44)	2.57 (1.44)	2.32 (1.44)	2.34 (1.44)

Table 1 (continued)

	Safety app group			Control group	Total
	TOT ($n = 117$)	Downloaded app ITT ($n = 106$)	Used app ITT ($n = 67$)	($n = 93$)	($N = 210$)
Limit or change activities	1.52 (0.94)	1.52 (0.96)	1.54 (0.96)	1.40 (0.80)	1.47 (0.88)
Keep a weapon at residence	1.99 (1.23)	1.96 (1.23)	1.99 (1.23)	1.91 (1.33)	1.95 (1.27)
Carry pepper spray/mace/weapon	2.30 (1.55)	2.26 (1.54)	2.48 (1.63)	2.38 (1.61)	2.33 (1.58)
Avoid nighttime classes	1.71 (1.11)	1.70 (1.12)	1.76 (1.19)	1.57 (1.03)	1.65 (1.08)
Endogenous variable pre-post difference scores					
Trust in mobile technology score	0.36 (0.81)	0.36 (0.81)	0.35 (0.72)	0.05 (0.86)	0.23 (0.84)
Negative consequences of apps	0.41 (1.22)	0.41 (1.21)	0.37 (1.15)	0.15 (1.27)	0.30 (1.24)
Cautious when using safety apps	0.38 (1.17)	0.37 (1.16)	0.42 (1.03)	− 0.03 (1.11)	0.20 (1.16)
Safety apps are risky	0.30 (0.87)	0.29 (0.90)	0.27 (0.96)	0.04 (0.99)	0.18 (0.93)
Fear of crime victimization score	0.11 (0.73)	0.06 (0.71)	0.06 (0.75)	0.12 (0.68)	0.12 (0.71)
Theft—day	0.15 (0.95)	0.12 (0.95)	0.15 (1.06)	0.17 (0.96)	0.16 (0.96)
Non-sexual assault—day	0.21 (0.96)	0.16 (0.96)	0.13 (1.01)	0.10 (0.83)	0.16 (0.91)
Sexual assault—day	0.18 (0.94)	0.13 (0.94)	0.09 (1.11)	0.11 (0.89)	0.15 (0.92)
Theft—night	0.10 (1.16)	0.04 (1.10)	0.01 (1.05)	0.06 (1.08)	0.09 (1.12)
Non-sexual assault—night	0.05 (1.03)	− 0.01 (1.01)	0.03 (1.00)	0.13 (0.91)	0.09 (0.98)
Sexual assault—night	− 0.01 (1.00)	− 0.08 (0.95)	− 0.07 (1.03)	0.16 (0.99)	0.07 (1.00)
Bystander intentions score	0.06 (0.50)	0.06 (0.52)	0.00 (0.51)	− 0.07 (0.47)	0.00 (0.49)
Watching a friend's belongings	0.09 (0.76)	0.08 (0.78)	− 0.02 (0.61)	− 0.01 (0.82)	0.05 (0.79)
Ensuring friend's safety after drinking	0.01 (0.61)	0.02 (0.60)	− 0.01 (0.71)	− 0.20 (0.61)	− 0.08 (0.62)
Walk with friend	− 0.06 (0.96)	− 0.05 (1.00)	− 0.06 (0.99)	0.05 (0.79)	− 0.01 (0.89)
Text/message friend	0.07 (0.82)	0.07 (0.85)	0.12 (0.83)	− 0.10 (1.07)	− 0.01 (0.94)
Watching a student's belongings	0.04 (1.03)	0.06 (1.06)	− 0.15 (0.94)	− 0.16 (1.09)	− 0.05 (1.06)

Table 1 (continued)

	Safety app group			Control group	Total
	TOT (n = 117)	Downloaded app ITT (n = 106)	Used app ITT (n = 67)	(n = 93)	(N = 210)
Ensuring student's safety after drinking	0.04 (0.88)	0.06 (0.87)	− 0.01 (0.88)	− 0.04 (0.88)	0.00 (0.88)
Walk with student	0.20 (1.12)	0.22 (1.13)	0.15 (1.03)	− 0.03 (1.02)	0.10 (1.08)
Self-protective behavior score	0.11 (0.50)	0.13 (0.50)	0.12 (0.51)	− 0.06 (0.66)	0.03 (0.58)
Avoid going outside alone at night	0.06 (1.19)	0.06 (1.24)	− 0.18 (1.19)	− 0.24 (1.46)	− 0.08 (1.32)
Limit or change activities	0.08 (0.75)	0.09 (0.77)	0.16 (0.75)	− 0.05 (0.85)	0.02 (0.80)
Keep a weapon at residence	0.13 (1.14)	0.21 (1.13)	0.38 (1.13)	0.85 (0.14)	0.14 (1.01)
Carry pepper spray/mace/weapon	0.09 (0.87)	0.15 (0.81)	0.05 (0.73)	− 0.24 (1.14)	− 0.06 (1.01)
Avoid nighttime classes	0.17 (0.85)	0.15 (0.88)	0.19 (0.84)	0.09 (0.99)	0.13 (0.91)

Measures

Pre- and post-surveys included demographic items as well as items measuring the exogenous and endogenous variables referenced in our hypotheses. The pre-survey asked whether participants had previously downloaded Rave Guardian. To avoid sensitizing participants to the purpose of this study, we asked participants to indicate whether they had ever downloaded each of ten distinct mobile apps, with Rave Guardian being one of these. Participants who indicated that they had previously downloaded Rave Guardian were allowed to complete both the pre- and post-survey but were excluded from our analysis.

As a measure of self-reported treatment compliance, we included a post-survey item asking treatment group participants whether they downloaded the Rave Guardian app, as instructed during the pre-survey. We also included two post-survey items asking treatment group participants how many times and how many hours, respectively, they used Rave Guardian. The percentage of treatment group participants who reported not using the app was substantial (with perfect overlap between the two items). Thus, we created a combined, dichotomized measure distinguishing those participants who reported not using the app (i.e., responded "I did not use the Rave Guardian app" to both items) from those who reported app usage.

Demographic survey items included gender (collapsed into cisgender woman, cisgender man, and transgender/gender non-binary), sexual identity (collapsed into straight and sexual minority), race (collapsed into White, Black, Asian, and Multiracial), Latinx ethnicity (0 = no; 1 = yes), year in school (freshman, sophomore, junior, senior), and live on campus (0 = no; 1 = yes). The exogenous variable indicated whether or not the participant was randomized into the safety app condition (0 = no; 1 = yes). Endogenous variables included trust in mobile technology, fear of crime victimization, self-protective behavior, and bystander intentions. We also asked participants whether they had experienced three forms of crime victimization within the past 30 days (i.e., non-sexual assault, sexual assault, and theft). However, the low number of participants who reported experiencing these outcomes during the 30-day pre- to post-survey interval precluded us from including them in our analysis (and, thus, we did not include these outcomes in our study hypotheses). Measures of endogenous variables that we analyzed in the study are described below.

Trust in technology We measured trust in technology with the 3-item "perceived risk" subscale of the Human-Computer Trust Scale (Gulati et al. 2019). This scale may be used to measure trust in many forms of technology, as each item includes a placeholder where researchers may specify a specific technology of interest. We inserted "mobile safety apps" into the placeholder of each item. Thus, the three items that we used to measure trust in technology are as follows: "I believe there could be negative consequences when using mobile safety apps," "I feel I must be cautious when using mobile safety apps," and "It is risky to interact with mobile safety apps." Participants rated these items on a 5-point Likert scale, ranging from "strongly disagree" (1) to "strongly agree" (5). We reverse coded these items and calculated scores by summing them and dividing by three. Cronbach's alpha in our sample was .76, with higher scores representing more trust (less perceived risk) in mobile safety apps.

Fear of crime victimization We measured fear of crime victimization with minor modifications to items used by Fisher and Sloan (2003) and Jennings et al. (2007) to measure fear of various forms of crime both during the day and night among college students. We asked participants to rate their agreement with a total of six versions of the following: "In the past thirty days, I have felt afraid that I might become a victim of [theft/non-sexual assault/sexual assault] during the [day/night]." Participants rated each of six items on a 5-point Likert scale, ranging from "strongly disagree" (1) to "strongly agree" (5). We calculated scores by summing them and dividing by six. Cronbach's alpha in our sample was .88, with higher scores representing greater fear of crime victimization.

Self-protective behavior We measured self-protective behavior with items used by Fisher and Sloan (2003) and Jennings et al. (2007) to measure constrained behavior in their studies of campus crime. We presented participants with five items asking whether concern about crime has led them to (1) avoid going outside alone at night, (2) limit or change activities, (3) keep a weapon in their place of residence for protection, (4) carry pepper spray/mace/other weapons, or (5) avoid taking nighttime classes. Participants rated each of these items on a 5-point Likert scale, ranging from "strongly disagree" (1) to "strongly agree" (5). We calculated scores by summing them and dividing by five. Cronbach's alpha in our sample was .74, with higher scores representing greater self-protective behavior.

Bystander intentions Because most existing bystander scales focus on sexual assault and intimate partner violence (e.g., the Bystander Intent to Help Scale developed by Banyard et al. 2014), we developed our own set of items asking about intentions to intervene into a range of crime situations. We asked participants six questions regarding their intentions to help if they saw someone [friend/student] in three risky situations. Participants rated their likelihood of (1) watching the belongings that a [friend/student] left unattended, (2) offering to walk with a [friend/student] who is walking across campus alone at night, (3) and making sure a [friend/student] gets home safely after having too much to drink. We also asked participants of their intentions to ask a friend to text/message them when they get home if the friend told them they were walking alone at night. We did not ask participants about their likelihood to offer help to a student (non-friend) in this latter situation because participants would not necessarily have the ability to text with a non-friend. Participants rated each of these seven items on a 5-point Likert scale, ranging from "strongly disagree" (1) to "strongly agree" (5). We calculated scores by summing them and dividing by seven. Cronbach's alpha in our sample was .84, with higher scores representing greater intentions to intervene.

Data analysis

Rates of missing data for each survey item ranged from 0 to 5.71%, and we imputed missing data using an expectation maximization algorithm with the Amelia II package in R 4.0.4 (Honaker et al. 2011). After imputing missing data, we calculated both pre- and post-survey scale scores for each of the endogenous variables, as described above. We then calculated pre- to post-survey difference scores by subtracting each pre-survey scale score from its corresponding post-survey score. We used these pre-post difference scores, which represent the pre- to post-survey increase in each endogenous variable, in our statistical analysis.

We used an intent to treat (ITT) approach in our main analysis by comparing participants assigned to the treatment group to those assigned to the control group, regardless of whether treatment group participants reported downloading or using the safety app. We also performed sensitivity analyses assessing the effect of treatment on the treated (TOT). First, we compared participants who were assigned to the treatment group and reported downloading the safety app on the post-survey to participants assigned to the control group. Then, we compared participants who were assigned to the treatment group and reported using the safety app on the post-survey to participants assigned to the control group.

To test our hypotheses, we employed path analysis methods (Asher 1983). We estimated path coefficients using the lavaan package in R (Rosseel 2021). Although our assignment of participants to treatment and control groups was based on class, we did not use multilevel modeling in our analysis. This is because our sample only consisted of four classes/clusters, and small numbers of clusters can increase the risk of a type 1 error in multilevel modeling (Austin and Leckie 2018; Leyrat et al., 2018; Maas and Hox 2005; Nugent and Kleinman 2021). Thus, we applied Satterthwaite's (1946) degree-of-freedom correction in our analysis, as Leyart et al.'s (2018) simulation study indicates that this reduces the risk of a type 1 error in analyses of data with a small number of clusters.

Results

Main ITT analysis

We performed chi-squared tests and *t*-tests, respectively, to determine whether any demographic variables or pre-survey endogenous scale scores differed between the ITT treatment and control group (see Table1 for descriptive statistics). None of these variables differed significantly between groups, so we did not include them as controls in our model. Standardized path coefficients for the ITT model are presented in Fig. 2. Contrary to Hypothesis 1, the path from safety app to fear of crime was non-significant, but consistent with this hypothesis, the path from fear of crime to self-protective behavior was significant and positive. The direct path from safety app to self-protective behavior was significant and positive, which is the opposite direction of the effect that we predicted in Hypothesis 2.

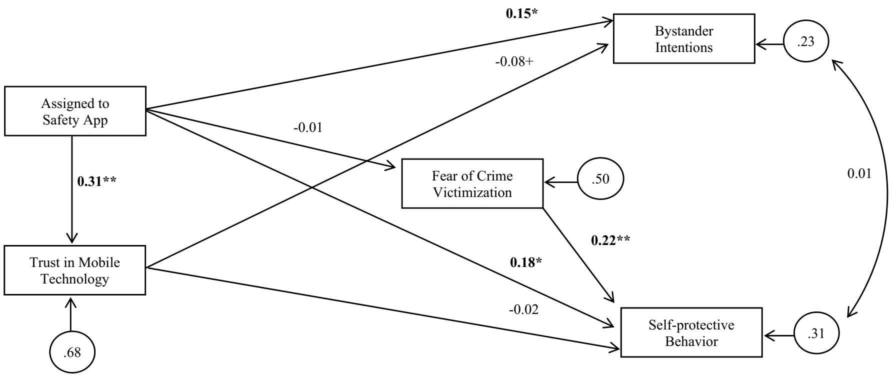

Fig. 2 ITT model standardized path coefficients for the relationships between safety app, trust in mobile technology, and outcomes ($N = 210$). Note: $+p < .10$, $*p < .05$, $**p < .01$, $***p < .001$. TLI = 1.00; CFI = 1.00; RMSEA = .00; SRMR = .02. Significant effects appear in bold

The direct path from safety app to bystander intentions was significant and positive, which is the opposite direction of the effect that we predicted in Hypothesis 3. Consistent with Hypothesis 4, the path from safety app to trust in technology was significant and positive, but contrary to what we expected, the path from trust in technology to bystander intentions was negative and non-significant (although marginal). Also contrary to Hypothesis 4, the path from trust in technology to self-protective behavior was non-significant.

The ITT model fit the data well, according to Hu and Bentler's (1999) standards, which recommend the following thresholds: TLI > .95; CFI > .95; SRMR < .08; RMSEA < .06. As noted at the bottom of Fig. 2, the fit measures for our model were as follows: TLI = 1.00; CFI = 1.00; RMSEA = .00; SRMR = .02. As indicated by the residual variance for each endogenous variable, the model explained 32% of the variance in trust in technology, 50% of the variance in fear of crime victimization, 69% of the variance in self-protective behavior, and 77% of the variance in bystander intentions.

The direct effect of safety app on bystander intentions was 0.15, which is small. The indirect effect of safety app on bystander intentions, running through trust in technology, was non-significant. Thus, the total effect of safety app on bystander intentions was 0.15, meaning that individuals assigned to the treatment group exhibited bystander intentions that were 0.15 standard deviations greater than those exhibited by the control group. This indicates that being assigned to the safety app condition was associated with greater bystander intentions, but this was not related to trust in technology.

The direct effect of safety app on self-protective behavior was 0.18. There were no significant indirect effects of safety app on self-protective behavior, neither through trust in technology nor fear of crime. Thus, the total effect of safety app on self-protective behavior was direct and small.

Sensitivity TOT analyses

In addition to our ITT analysis, we also conducted two TOT analyses. First, we dropped treatment group participants who reported that they did not download the safety app. Self-reported treatment compliance was high (90.60% of the treatment group). Thus, the sample size for the "downloaded app" TOT (N = 199) analysis was not much smaller than that of the ITT analysis (N = 210). Chi-squared tests of independence and t-tests, respectively, indicated that the "downloaded app" TOT treatment group and control group did not significantly differ ($p > .05$) on any demographic variables or pre-survey endogenous scale scores (see Table 1 for descriptive statistics). Findings from the "downloaded app" TOT analysis, summarized in Fig. 3, were substantively similar to those of the ITT analysis. Specifically, the significance of effects did not change, and the magnitude of coefficients only changed minimally such that the direct effect of safety app on bystander intentions was 0.16 (versus 0.15 in the ITT model) and the direct effect on self-protective behavior was 0.22 (versus 0.18 in the ITT model). Like the ITT model, there were no significant indirect effects of safety app on bystander intentions or self-protective behavior.

The "downloaded app" TOT model fit the data well, according to Hu and Bentler's (1999) standards: TLI = 1.00; CFI = 1.00; RMSEA = .00; SRMR = .03. As indicated by the residual variance for each endogenous variable, the model explained 32% of the variance in trust in technology, 52% of the variance in fear of crime victimization, 69% of the variance in self-protective behavior, and 76% of the variance in bystander intentions.

For our second TOT analysis, we dropped treatment group participants who reported that they did not use the safety app (i.e., responded "I did not use the Rave Guardian app" to post-survey items asking about number of hours and number of times respondents used the app). Self-reported treatment compliance with using the app was 63.21% among the treatment group, yielding a sample of 160. Chi-squared tests of independence and t-tests, respectively, indicated that the

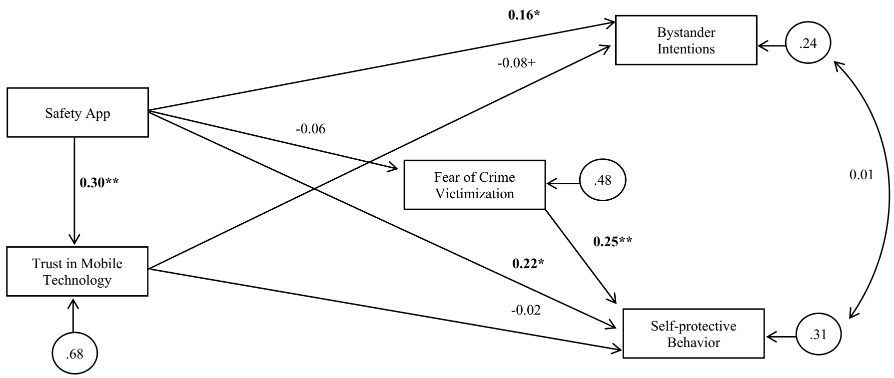

Fig. 3 "Downloaded app" TOT model standardized path coefficients for the relationships between safety app, trust in mobile technology, and outcomes (N = 199). Note: +p < .10, *p < .05, **p < .01, ***p < .001. TLI = 1.00; CFI = 1.00; RMSEA = .00; SRMR = .03. Significant effects appear in bold

"used app" TOT treatment group and control group did not significantly differ ($p > .05$) on any demographic variables or pre-survey endogenous scale scores (see Table 1 for descriptive statistics).

Findings from this "used app" TOT analysis, summarized in Fig. 4, differed from the ITT analysis in meaningful ways. The most substantively important differences are that the direct effect of safety app on bystander intentions was significant and positive in the ITT analysis, but non-significant in this TOT analysis. Thus, whereas participants assigned to the treatment group had significantly greater pre- to post-survey increases in bystander intentions than the control group, the subsample of treatment group participants who reported using the safety app did not significantly differ from control group participants on bystander intentions.

Additionally, while there was a non-significant indirect effect of safety app on bystander intentions in the ITT model, there was a significant, negative indirect effect in the "used app" TOT analysis. In this TOT model, safety app had a significant, positive effect on trust in mobile technology, which, in turn, had a significant, but very small negative effect on bystander intentions. This effect decomposes to − 0.04 [i.e., (.30)*(− .13)]. Thus, whereas the total effect of safety app on bystander intentions was small and positive in the ITT analysis (via one direct effect), it was very small and negative in the "used app" TOT analysis (via an indirect effect that runs through trust in mobile technology). The effect of safety app on self-protective behavior was substantively similar between the ITT and the "used app" TOT analyses (small and positive).

The "used app" TOT model fit the data well, according to Hu and Bentler's (1999) standards: TLI = 1.00; CFI = 1.00; RMSEA = .00; SRMR = .03. As indicated by the residual variance for each endogenous variable, the model explained 36% of the variance in trust in technology, 50% of the variance in fear of crime victimization, 67% of the variance in self-protective behavior, and 78% of the variance in bystander intentions.

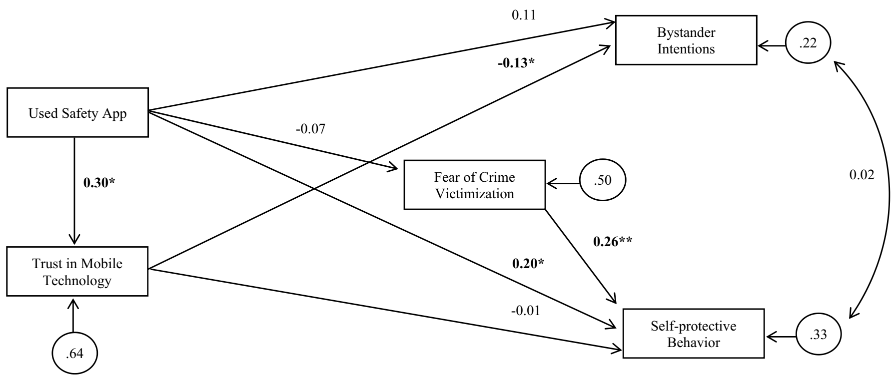

Fig. 4 "Used app" TOT model standardized path coefficients for the relationships between safety app, trust in mobile technology, and outcomes ($N = 160$). Note: +$p < .10$, *$p < .05$, **$p < .01$, ***$p < .001$. TLI = 1.00; CFI = 1.00; RMSEA = .00; SRMR = .03. Significant effects appear in bold

Discussion

The differences between our ITT analysis and our “used app” TOT analysis point to some important implications for understanding the impact of mobile safety apps on crime prevention behavior. Specifically, our findings suggest that there may be different effects of accessing versus using such apps.

Contrary to expectations, being randomized to the mobile safety app condition was associated with significantly higher levels of bystander intentions and self-protective behavior. Thus, being assigned to the mobile safety app condition did not appear to amplify a bystander effect that diffuses responsibility for helping individuals in emergency situations, as proposed in previous scholarship (see Kettrey and Thompson 2022). Instead, it was associated with increased intentions to help. According to Latane and Darley (1969), bystanders must meet four criteria to intervene in an emergency situation: notice the event, define the event as warranting action/intervention, take responsibility for acting, and demonstrate a sufficient level of self-efficacy to intervene. We hypothesized that mobile safety app access would inhibit study participants from taking responsibility for acting. This was based on the assumption that participants would believe people in danger of crime victimization could use mobile devices/apps to contact a wide range of people for help. Instead, findings suggest that mobile safety app access is associated with greater responsibility (or intentions) to act. It is possible that access to a mobile safety app may sensitize participants to crime threats (leading them to notice risk factors for crime and/or define them as warranting action) or make participants feel better equipped with technological tools to handle crime threats (increasing self-efficacy to act). The fact that the safety app includes functions that allow users to report potential crime may suggest to users that it is their responsibility to take some sort of action to prevent criminal activity and may increase their self-efficacy to do so.

It is also noteworthy that, contrary to expectations, our ITT findings indicate that being randomized to the mobile safety app condition was associated with significantly higher levels of self-protective behavior. Based on the proposition that mobile safety apps may decrease fear of crime (Bivens and Hasinoff 2018; Hasinoff 2017; Maxwell et al., 2020), along with past research linking fear of crime to safer behaviors (Liska et al. 1988; Rader et al. 2009), we hypothesized that being randomized to the mobile safety app condition would lead to less self-protective behavior. Yet, our findings indicate that access to mobile safety apps did not influence fear of crime, but it did increase self-protective behavior. Similar to our explanation for our observed effects on bystander intentions, it is possible that access to a mobile safety app sensitizes users to crime, albeit without increasing fear. Future research should examine the effect of mobile safety app access on participants’ awareness of crime threats/activities in their communities.

Interestingly, our “used app” TOT analysis indicated that treatment group participants who reported actually using the mobile safety app exhibited greater increases in self-protective behavior than those in the control group (similar to the ITT analysis), but did not exhibit greater increases in bystander intentions. In fact, our hypothesis that the mobile safety app would be associated with greater trust in technology which would in turn be associated with decreased bystander intentions was supported in the "used app" TOT analysis (although this effect was very small) but not in the ITT analysis.

Comparing findings from the main analysis and the "used app" TOT analysis suggests that both accessing and using a mobile safety app are beneficial for self-protective behavior. However, accessing a safety app may be more beneficial for bystander intentions than actually using it. That is, as previously discussed, having access to a mobile safety app may make individuals more aware of crime and may sensitize them to the potential dangers that their friends and others may face. However, actually, using the app may foster a trust in technology that hinders bystander intentions to help others. Essentially, our hypothesis that trust in mobile technology would thwart bystander intentions may apply to app usage, but not app access, suggesting there may be a threshold of mobile app interaction that increases trust in technology and inhibits bystander intentions to help others. Future research that uses a larger sample and follows participants for a longer period of time should explore this possibility.

Limitations

Findings from our study must be interpreted within the confines of a few important limitations. First, although we believe that randomization ensured a reasonable degree of internal validity of our study, the geographic location and demographic composition of our sample threaten external validity. In other words, our findings do not necessarily generalize to populations that are different from the one we studied. This is important considering that the majority of participants in our sample were White, non-Latinx, straight, and first-year students. Second, our 1-month follow-up period only permitted us to determine effects of mobile safety apps during a circumscribed timing interval. We cannot speculate about the longer-term effects of mobile safety apps on our outcomes. Third, and relatedly, the fact that few participants reported experiencing crime victimization during our 1-month follow-up period precluded us from examining this outcome. Thus, future research should examine effects of mobile safety apps over a longer time period that would allow examination of crime victimization outcomes.

Conclusion

In this experimental study, we tested the general proposition that mobile safety apps function as a form of crime control theater, defined as "a public response or set of responses to crime which generate the appearance, but not the fact, of crime control" (Griffin and Miller 2008, p. 160). As DeVault et al. (2016) note, crime control theater is not only ineffective, but may also be associated with unintended consequences. In our study, we were specifically interested in whether mobile safety apps were associated with two unintended consequences that may be counter-productive to crime prevention: decreased bystander intentions and decreased self-protective behaviors.

Our findings suggest that mobile safety apps may serve as a form of crime control theater but only among those who actively use the app (not those who merely access it) and only for bystander intentions to help others who may be at risk of experiencing crime. Among those who access but do not use the app, we found a positive effect on bystander intentions to help potential victims of crime, as well as on self-protective behaviors that may reduce one's personal risk of crime. In this sense, our study does not provide evidence that mobile safety apps function as a form of crime control theater. However, for those who report using the app, this technology increased self-protective behavior but inhibited bystander intentions to help others. In this sense, mobile safety apps do appear to represent a form of crime control theater for a particular outcome and a particular subgroup. Taken together, our findings appear to suggest that app access is more helpful than app usage. Mobile safety app access may sensitize individuals to crime risk and promote crime prevention behavior, yet using the app may foster a dependence on technology that inhibits some crime prevention behaviors. Future research should aim to further explore this possibility such as by using a three-armed randomization design in which participants are assigned to (1) a treatment group instructed to download a mobile safety app (with analytical attention to those who use it versus those who do not), (2) a no-treatment control group, or (3) an attention control group receiving information expected to sensitize participants to crime risk. Until more research accumulates to elucidate the nuanced effects of mobile safety app access versus usage, widespread uptake of this technology appears to be premature.

Data Availability Data are available from the lead author, by request.

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Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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